

Original Article

Heat action plans in eight Indian cities: Knowledge gaps & opportunities for intersectoral heat governance

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Background & objectives: Extreme heat in cities poses significant health risks especially to vulnerable populations. Climate change has led to heatwaves in India increasing in frequency and duration. To address the heat-health challenges, heat action plans (HAPs) are a key instrument for heat governance. The objective of this study is to critically assess city level HAPs from India, in terms of the eight core elements identified by the World Health Organization guidance on heat-health action.

Methods: The study uses a benchmarking approach in mapping the representation of core elements across eight city-level HAPs, to assess the plan's coverage and extent of development for each element and its sub-elements. Supportive text analysis was conducted using R software to study the frequency of use of words in the HAPs and key words corresponding to core elements.

Results: The HAPs varied in design and scope, with core elements that require long-term institutional and/or intersectoral planning and implementation receiving least attention. These elements included care for vulnerable populations, health system preparedness, long-term urban planning, and surveillance of health outcomes. The study identifies the dominant framing of the issue of heat as a time-limited disaster event (namely, a heatwave) as a significant barrier in designing HAPs that are responsive to local contexts and calls for long-term measures required to shape structural drivers of differential vulnerabilities.

Interpretation & conclusions: A paradigm shift from solely top-down disaster management to coupling with decentralised, community-informed management and long-term measures is essential to effectively address heat-related health risks in both the immediate and the long-term.

Key words Cities - heat action plans - heat-health governance - heat-health risks - India - WHO

Climate change has elevated extreme heat from an 'invisible' public health threat to an urgent challenge, with heat stress being a leading cause of weather-related deaths, and disproportionately affecting warmer regions and the poor¹⁻³. Between

2001 and 2013, the population of India exposed to extremely hot temperatures increased by 10 million⁴. It is estimated that cities could warm by 4°Celsius on average by 2100, posing significant health challenges for populations living within dense urban networks⁵.

It is well-established that both hot temperatures and heatwaves are associated with excess mortality risk^{5,6}. A heatwave is usually defined as ‘a period of extreme high temperature that lasts several days’⁷. There is no internationally accepted definition of heatwaves. Definitions vary based on how a heatwave is identified and what indicators are used to measure its various dimensions^{6,8}. Beyond fatalities, extreme heat is also linked to the disruption of the everyday and the ordinary: school closures⁹, livelihood loss¹⁰, water shortages¹¹, power outages¹², and food insecurity¹³. These impacts on lives and livelihoods are further shaped by one’s ability to respond to heat a relationship mediated by an individual’s social, cultural, economic, and physiological status¹⁴. To address these heat-health risks and their differential impacts on vulnerable groups, heat action plans (HAPs) have emerged as a dominant planning instrument.

Under the aegis of the World Health Organization Regional Office for Europe (WHO/Europe), the EURO-HEAT Project (2005-2007) guidance document on heat-health action plans (HHAPs) was developed for designing national and sub-national HAPs¹⁵. The eight ‘core elements’ of a comprehensive HHAP include: (i) agreement on a lead body for coordination, (ii) accurate and timely alert systems, (iii) heat-related health information plan, (iv) reduction in indoor heat exposure, (v) particular care for vulnerable population groups, (vi) preparedness of the health and social care system, (vii) long-term urban planning, and (viii) real-time surveillance and evaluation¹⁶. City-specific plans generally differ based on how each core element is incorporated, the associated scope and complexity of implementation¹⁷.

A severe heatwave in Odisha in 1998 led to the development of the first State heat response plan in India¹⁸. Following the release of the WHO/Europe guidance document, more HAPs were launched in the country¹⁹. Several State governments and municipal corporations have received funding and external technical support from domestic and international organisations and experts to develop HAPs in India. Ahmedabad’s 2013 HAP was developed in collaboration with the Indian Institute of Public Health (IIPH) Gandhinagar, Public Health Foundation of India (PHFI), Natural Resources Defense Council (NRDC), Mount Sinai School of Medicine, and Rollins School of Public Health at Emory University. The Bhubaneswar HAP, launched in 2016, was prepared by Integrated Research and Action for Development (IRADe) and

supported by the International Development Research Center, Government of Canada. The 2019 Rajasthan HAP was developed in collaboration with UNICEF Rajasthan’s Disaster Risk Reduction Section²⁰. In an analysis of 37 HAPs, significant variations were found in the actions recommended to be carried out in response to heatwaves, ranging from health systems capacity- building to green roofs²¹.

The present study was conducted to critically assesses eight HAPs for Indian cities, in terms of their coverage of the eight core elements identified by WHO/Europe as crucial components for heat-health planning and action, benchmark them against global standard, and to identify gaps and opportunity on heat-health management through these programmes.

Materials & Methods

This observational study was undertaken by the Environmental and Resource Economics Unit, Institute of Economic Growth, Delhi, India between August 2024 and February 2025. All the data provided in this study is available in the public domain.

Study period: The data were collected from sources and documents in the public domain. No identifiable data of individuals or biological or environmental sampling was involved.

Analytical framework: The study assesses eight city-level HAPs in India, benchmarking the contents of the HAPs against the eight core elements of heat-health action planning identified in WHO/Europe’s guidance document for heat-health action planning. It further identifies and assesses the core elements by sub-elements adapting the indicator-based approach²², which linked sub-elements to the core elements developed under the EuroHEAT project. The document text corresponding to the core elements was also analysed using the text analysis method, to complement and validate the key findings.

The analysis was done in three stages. First, an iterative review of the selected HAPs was conducted to finalise city-specific sub-elements and identify key words for the textual analysis. Second, a colour-coded data extraction form was developed and completed, capturing the findings from the analysis of the HAPs. Third, a text analysis was conducted with the help of identified key words to complement the findings from the review. The researchers adopted a consensus-based

approach for the first and second stages. The details are presented below.

An iterative protocol was developed for an initial review of the HAPs, to build familiarity with their structure and content. Four members (RD, PD, AM, GS) in the research team reviewed the HAPs independently and iteratively until all members agreed on the best representation of the data. The team members collaborated on the principle of ‘mutual respect, equal involvement, and shared power’^{23,24}. This led to modifications in the sub-elements to reflect city-level specificities, since originally²² these were intended for characterizing national level plans. The sub-element ‘inclusion in national disaster preparedness’ was changed to ‘inclusion in disaster preparedness,’ ‘cross-border cooperations’ was changed to ‘synchronicity with district- and State- level plans’ and ‘involving data from >1 region/city’ was changed to ‘involving data from >1 ward’, since wards are the administrative divisions within a city. ‘Changes to water supply management’ was added as a sub-element for the core element of ‘long-term urban planning’, as water supplies are critical for managing extreme heat conditions in Indian cities. Table I provides an overview of the core elements, their sub-elements and key words identified for text analysis.

Three distinct classifications for the characterisation of content on elements and sub-elements emerged from the review. A colour-coded labelling system was accordingly developed to reflect these categories depending on whether there was (i) no mention of the sub-element (light blue), (ii) mention without any actionable details and/or guidance for implementation (dark blue) and (iii) mention that is elaborated upon with actionable details and/or guidance for implementation (navy blue). The no mention category includes instances where a key word from the sub-element is used without supportive text to develop the sub-element. Data were extracted to a form designed in Microsoft Office Excel. Subsequently, a sub-set of two researchers independently reviewed each HAP to classify its contents in terms of the sub-elements represented in it (or its lack thereof). This action was repeated over two rounds to confirm agreement (or lack of it where it persisted). Feedback sessions were held with the larger research team after each round and inputs incorporated. Consensus was reached on the labelling attributed to each element and sub-elements by the end of the second round.

The researchers drew upon their careful reading of the documents in the first two stages and the enumeration of relevant keywords to conduct a text-analysis²⁵. Text analysis was conducted using R software to study the frequency of use of words in the HAPs and key words corresponding to the core elements. The ten most frequently used words were identified for each of the HAPs, expressed as a percentage of the total number of words in the respective HAP to probe on the variability of usage. Key words were identified for each core element in the WHO/Euro guidance. The frequency of these were then examined in each HAP using text analysis. Text analysis is employed as a complementary tool to the primary benchmarking process, rather than as a standalone method for determining word significance. An iterative approach was adopted, involving repeated cycles of text analysis and document review. The soft copies of the HAP were searched with the keywords, so that we could see the sentences, paragraphs and sections within which these keywords were used. This ensured that the contextual use of keywords was assessed for relevance to specific sub-themes, further validating the findings.

Selection of HAPs: The national government is working with several States, districts and cities in developing HAPs. A recent estimate suggests that there are over 130 cities and districts involved in the process²⁶. Most HAPs are at different stages of development and revisions, and hence unavailable in the public domain or in one database. For the present study, publicly accessible city-level HAPs at the time of the study were accessed using a desktop search-engine based strategy (Google search engine) along with reviewing previous studies and grey literature regarding HAP development in India^{21,27}. For the analysis, two criteria were considered for selection of HAPs. Firstly, the HAP had to be titled as either a ‘heat action plan’, ‘heatwave action plan’ or specifically mention in the title page that it addresses extreme heat or heatwave response planning. Secondly, it was also required that the HAP be an official policy document, owned by the concerned city administration and approved for implementation.

Out of 10 city-level HAPs that met these criteria at the time of the analysis, eight could be fully accessed. Two HAPs (for the cities of Gondia and Chandrapur) could not be accessed fully and were therefore not considered for the analysis. The issuing authority, partners of the HAPs and sources of HAPs are presented as supplementary table. It is to be noted that

Table I. Core elements, sub-elements, and key words for HAP analysis

Core element	Sub-elements	Key words identified for text analysis
Agreement on a lead body and clear definition of actors' responsibilities	Clearly defined lead body Governance structure Involvement of >1 other agencies Regular meetings and/or reviews Inclusion in disaster preparedness Synchronicity with district and State-level plans	Department, prepare, nodal, IMD, Government, disaster, review, stakeholders, review, roles, District, corporation, NDMA, authority, meeting, ministry
Accurate & timely alert systems	Threshold definition scientifically sound Regionally adapted definitions Warning is issued well in advance Different alert levels for different levels of action Alert is communicated following a clear plan	Alert, IMD, forecast, waves, communication, threshold, temperature, meteorological, emergency
Health information plan	Clearly defined actors/recipients/contents Effective dissemination of information (>1 channel) Quality of advice Public & professionals addressed Appropriate timing of information campaign	Paramedics, medical, wards, doctor, hospitals, information, campaign, healthcare, patients, centres, advice, nurses, awareness, prevent, readiness
Reduction in indoor heat exposure	Giving advice Providing cool rooms/spaces Provision or use of mobile coolers Planning or support for increased albedo or shading Planning or support for better insulation	Shelters, sheds, insulation, ventilation, thermal, island, indoor, cool, air conditioner, fans, building, home, coolers, albedo, shade, walls, roof, paint, conditioners
Particular care for vulnerable groups	Identification of relevant groups (>1) Activation of a telephone service Specific measures (buddies, neighbours) Regular re-assessment of vulnerable population groups Information and training for caregivers	Vulnerable, group, slum, elderly, children, pregnant, disabled, homeless, income, women, older, disabilities, chronic, community
Preparedness of the health/social care system	Increase of capacity of health services Heat reduction in health care facilities Special precautions in nursing homes Special resources for patients/public Improving health-care networks	Paramedics, medical, wards, doctor, hospitals, information, campaign, healthcare, patients, centres, advice, nurses, awareness, prevent, readiness
Long-term urban planning	Increased green & blue spaces Changes in building design (albedo, insulation, passive cooling) Changes in land-use decisions Energy consumption reduction Changes in water supply management Individual & public transport policies	Green, spaces, shade, trees, building, energy, water, land, land-use, planning, long, long-term, electricity
Real-time surveillance	Less than 48-h interval Involving data from >1 ward Involving data from >1 health effect Use for adjustment of measures Use for evaluation of effectiveness	Surveillance, data, reviewing, evaluate, effectiveness, tracking, indicator, indicators, database, record, monitor, review, monitoring, evaluation

Source: Adapted from Ref 22

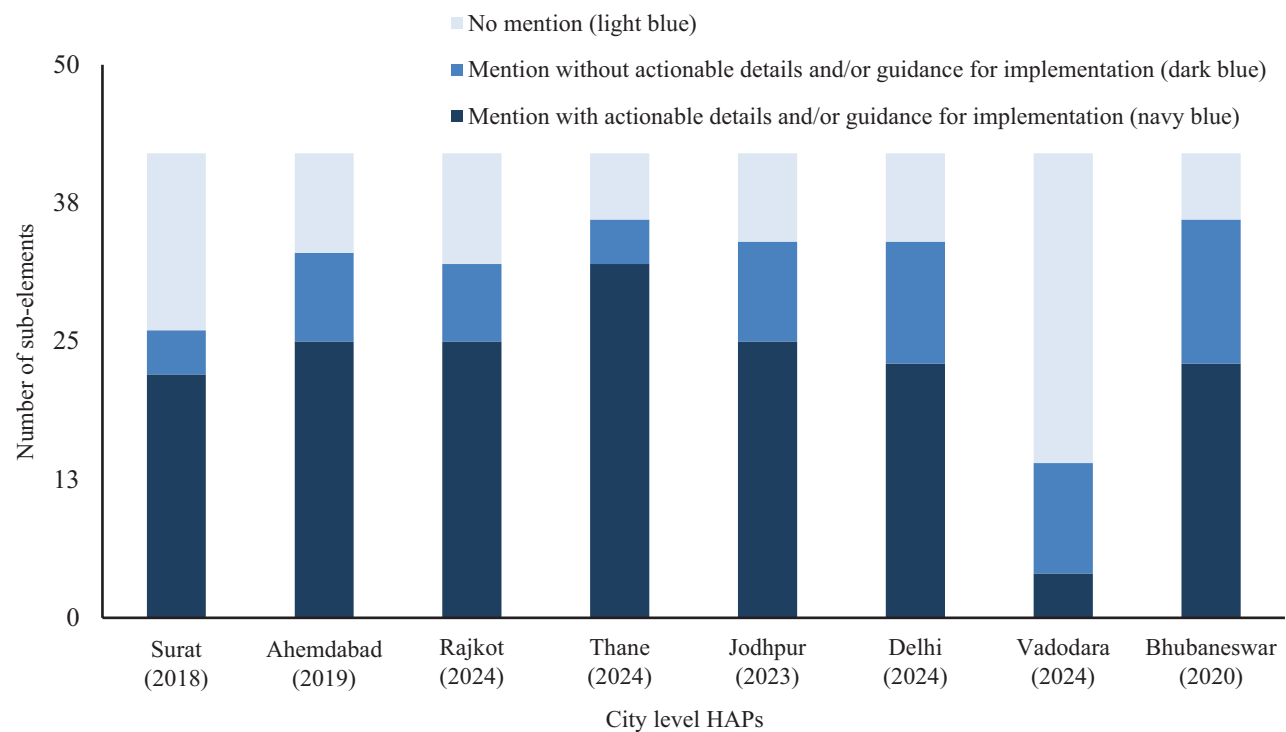


Fig. 1. Overview of sub-elements across the eight city-level HAPs.

the selection process is internally consistent in keeping with the objective, which is to examine the presence of different components in each individual HAP.

Results

Figure 1 presents coverage of the sub-elements across the HAPs, by the color-coded categorisation. Thane has the highest coverage of sub-elements at 38, followed by Bhubaneswar (37) and Delhi (37). Vadodara has the least coverage at 17, followed by Surat (28). The Thane HAP also has the least number of sub-elements labelled ‘dark blue’ and ‘light blue’, indicating that the HAP is most likely to be successful in aligning policy objectives with the required guidance amongst the studied ones.

The element ‘health information plan’ has the best coverage amongst the core elements, followed by ‘agreement on a lead body’ and ‘accurate and timely alert systems.’ The core elements that require identifying and addressing long-term and/or contextual factors (‘particular care for vulnerable groups,’ ‘real-time surveillance,’ and ‘long-term urban planning’) are least represented.

The text analysis also provides an overview in terms of the use of keywords. Figure 2 shows the top ten words used in each of the eight HAPs. Word frequencies are presented as a percentage of total

words within each document, allowing for relative comparisons both within and across documents.

‘Heat’ is expectedly the most used word across HAPs, followed by ‘wave’. ‘Health’ is the fourth or fifth most used word. Temperature, risk and stress show high variability in usage, finding a place in the top ten words in five, two and two HAPs, respectively. The Ahmedabad HAP’s use of ‘cool’ and ‘roofs’ emphasises its focus on this adaptation measure among other words, and this usage was also distinctly higher compared to the HAPs for other cities. ‘Water’ is mentioned frequently too, and features in the top ten words used in five HAPs, signifying its importance for heat management.

The findings on the core elements are discussed below. Tables IIA and IIB present the color-coded results on sub-elements. A graphical representation is also provided in supplementary material.

Agreement on a lead body and clear definition of actors’ responsibilities: The responsibility for HAPs is vested with the National Disaster Management Authority (NDMA) at the national level. For cities, the corresponding State disaster management authority leads heat governance. The text analysis confirms the significance of the India Meteorological Department (IMD) and the NDMA. It also demonstrates variability in the frequency for some key words, such as ‘disaster’,

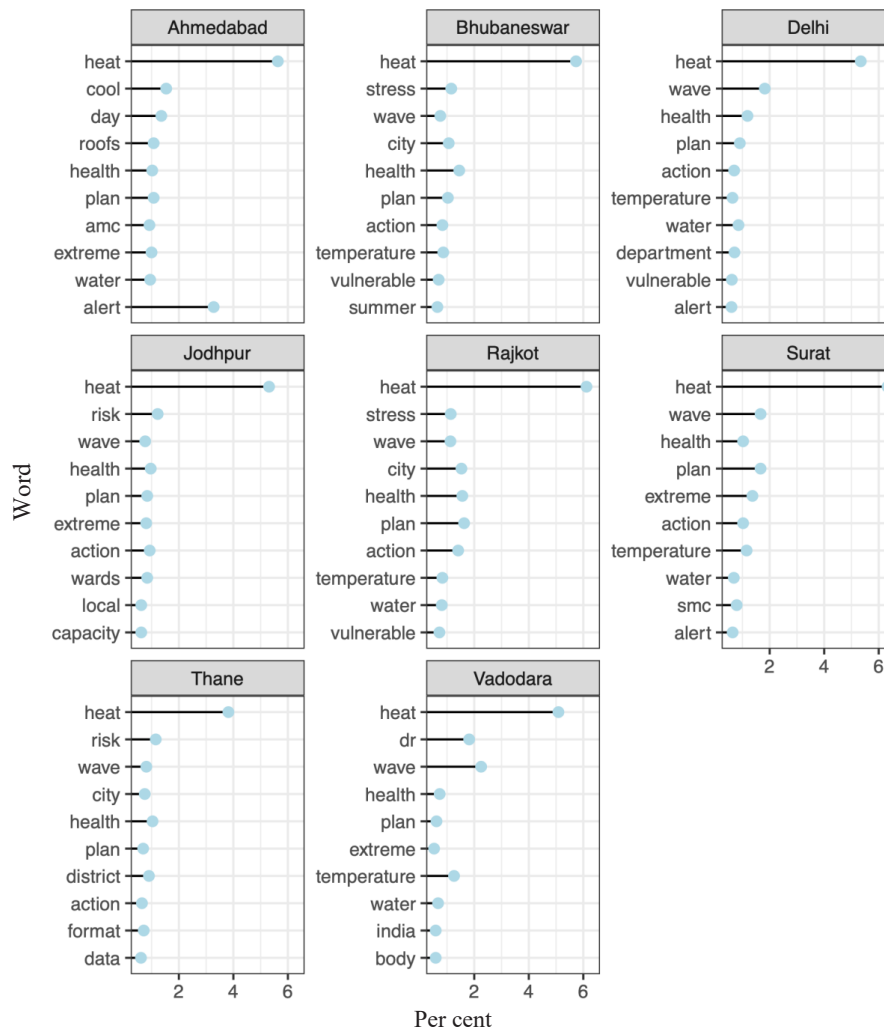


Fig. 2. Ten most frequently used words in each HAP (as a percentage of total words), generated with the tidy text package, R software.

where usage ranges from zero occurrence (Ahmedabad) to very high occurrence (Vadodara). Emphasis on the designated lead body (Delhi) and on assigned roles can be strengthened (Vadodara). The Delhi HAP identified relevant actors and assigned responsibilities. Some HAPs mandated the establishment of committees for implementation (Heatwave Task Force, Thane Municipal Corporation; Heatwave Plan Implementation Committee, Jodhpur; Steering Committee for Medical Emergency Preparedness, Bhubaneswar). Surat and Thane cover all the sub-elements. Alignment with district- and State-level institutional mechanisms and plans falls short in all the others. Three HAPs mention inclusion and disaster preparedness, but lack actionable guidance.

Accurate and timely alert systems: All HAPs perform well on this element, though Vadodara has scope for

significant improvement. The Ahmedabad, Rajkot and Bhubaneswar HAPs cover key aspects of heat-health alert systems (scientifically sound threshold definition, regionally adapted definition, advanced warning, graded level of alerts and communication plan). Notably, five HAPs are strong in having different levels of actions for different levels of alerts. For some HAPs, communication plans need improvement (*e.g.*, Delhi HAP). IMD definitions for heatwaves prevail across HAPs, with some variations in the definition of heat thresholds adapted to local contexts. For instance, the Thane HAP provides locally determined felt temperature thresholds based on the heat index to indicate the impact of humidity in addition to dry temperature. The text analysis confirms high levels of similarity in the usage and frequency pattern of key words.

Table IIA. Results from the colour-coding exercise on sub-elements (1-4)

Element / Sub-element City	1. Agreement on a lead body and clear definition of actors' responsibilities					
	Clearly defined lead body	Governance structure	Involvement of >1 other agencies	Regular meetings and/or reviews	Inclusion in disaster preparedness	Synchronicity with district- & State-level HAPs
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						
2. Accurate & timely alert systems						
	Threshold definition scientifically sound	Regionally adapted definitions	Warning is issued well in advance	Different alert levels for different levels of action	Alert is communicated following a clear plan	
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						
3. Health information plan						
	Clearly defined actors/ recipients/contents	Effective dissemination of information (>1 channel)	Quality of advice	Public & professionals addressed	Appropriate timing of information campaign	
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						
4. Reduction in indoor heat exposure						
	Giving advice	Providing cool rooms/spaces	Provision or use of cooling equipment	Planning or support for increased albedo or shading	Planning or support for better insulation	
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						

Light blue: no mention; Dark blue: mention without actionable details and/or guidance for implementation; Navy blue: mention with actionable details and/or guidance for implementation

Table IIB. HAPs: Results from the colour coding exercise on sub-elements (5-8)

Element/Sub-element City	5. Particular care for vulnerable groups					
	Identification of relevant groups (>1)	Activation of a telephone service	Specific measures (buddies, neighbours...)	Regular re-assessment of vulnerable population groups	Information & training for caregivers	
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						
6. Preparedness of the health/social care system						
	Increase of capacity of health services	Heat reduction in health facilities	Special precautions in nursing homes	Special resources for people/public	Improving healthcare networks	
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						
7. Long-term urban planning						
	Increased green & blue spaces	Changes in building design (albedo, insulation, passive cooling)	Changes in land-use decisions	Energy consumption reduction	Changes in water supply management	Individual & public transport policies
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						
Bhubaneswar						
8. Real-time surveillance						
	Less than 48-h interval	Involving data from >1 ward	Involving data from >1 health effect	Use for adjustment of measures	Use for evaluation of effectiveness	
Surat						
Ahmedabad						
Rajkot						
Thane						
Jodhpur						
Delhi						
Vadodara						

Light blue: no mention; Dark blue: mention without actionable details and/or guidance for implementation; Navy blue: mention with actionable details and/or guidance for implementation

Box. Gender in the Bhubaneswar HAP

The Bhubaneswar HAP (2020) provides specific actionable guidance linked to gender-based vulnerability. It recommends creating and implementing gender-specific heat-health guidelines by adopting heat-focused examination procedures at local hospitals and increasing gender-sensitive outreach in slum settlements. In the pre-heat season, the plan recommends workshops for link and frontline workers to support gender-sensitive outreach and community surveillance. During the heat season, the planned activities include gender-sensitive medical examinations at local hospitals and urban health centres, involving frontline health workers such as ASHAs and Anganwadi workers. The plan also calls for adapting pharmacological treatments with 'gender aspects.' In the post-heat season, it advises reporting of gender-based surveillance data for evaluation and informed decision-making.

Source: Bhubaneswar Heat Wave Action Plan 2020

Health information plans: 'Health information plans' was the best-represented core element with all sub-elements being mentioned in every HAP. Information was generally geared towards medical professionals. Ahmedabad and Jodhpur have scope to improve on quality of advice with more guidance, and Vadodara in terms of information campaigns. In terms of frequency of specific key words, the five HAPs which excel in performance are highly similar. Relatively higher variation is seen in frequency of key words amongst Ahmedabad, Vadodara and Jodhpur.

Reduction in indoor heat exposure: Among the sub-elements, insulation is not covered well in 4 HAPs. There is scope for improvement in providing advice on tackling exposure in the Vadodara HAP. Except for one HAP, improved albedo/shading is well covered. Specific programmes for cool roofs and cool spaces using white reflective paint are also included. On the sub-theme of provision or use of cooling equipment all HAPs mention fans and/or coolers. The Rajkot HAP mentions rise in interpersonal violence due to rising temperatures. Bhubaneswar, Thane, Delhi and Rajkot excel in this core element. Findings from the text analysis are complementary, displaying high levels of similarity in usage and frequency of key words, in these four HAPs.

Caring for vulnerable groups: Identification of vulnerable communities is a key pre-requisite for public health interventions. The HAPs tend to fall short in this aspect. The needs of vulnerable groups are dynamic, and regular re-assessments are required alongside ongoing monitoring for active responses.

The HAPs lacked mention of such periodic assessments and follow ups. Methodologies or protocols to identify vulnerable groups are required. Four HAPs provide information on why particular groups are vulnerable to heat-health risks. Thane and the Bhubaneswar conducted vulnerability assessments to identify vulnerable city wards in the formative phase. The Rajkot HAP mentions delivering targeted interventions for vulnerable groups. The Bhubaneswar plan specifically focuses on gender (Box). Textual analysis reveals less/no mention of 'disabled' and 'homeless' across the HAPs, while 'children' and 'elderly' are well covered by nearly all the HAPs.

Preparedness of health/social care systems: The HAPs lack information on at least one of the sub-elements under this core element. Two HAPs make mention of 'heat reduction in health facilities' and 'special precautions in nursing homes.' Suggested interventions include the setting up of air-conditioned wards and cool wards (Delhi, Bhubaneswar). Interventions for health care institutions include provision of drinking water (Ahmedabad) and organizing workshops for staff members and health care professionals (Rajkot). The other sub-elements ('increase in capacity of health services, 'special resources for people/public', 'improving healthcare networks') are well represented across HAPs, albeit with heterogeneity in terms of the actions covered under them.

Long-term urban planning: Long-term measures are poorly represented in most HAPs, except for Thane and Delhi. While 'increase in green and blue spaces' and 'changes in building design' are covered in most HAPs, interventions that require more complex decision-making by public authorities and are shaped more explicitly by political factors such as 'changes in land-use decisions' are less represented. The Thane HAP, highlights heat risk informed urban planning for mitigating future heat risks.

Real-time surveillance and evaluation: Surveillance data is required for the monitoring of health impacts of heatwaves and to inform decision-makers in evaluating and selecting HAP measures. While six HAPs had a component on surveillance, the use of surveillance data for dynamic adjustment of measures found mention in three HAPs. The use of surveillance data for the evaluation of the effectiveness of the HAP finds mention in all plans. Thane provided a monitoring and evaluation framework to carry out evaluation

activities on an annual basis. Thane, Jodhpur and Delhi have relatively better coverage, especially in ‘use for adjustment of measures’ sub-element, which has no mention in the other HAPs. However, there is scope to improve and strengthen guidance across all HAPs.

Discussion

This analysis has unpacked the constituent core elements and sub-elements for eight HAPs in India. Each city HAP presents with strengths and weaknesses across elements, all of which are analysed without any subjective judgement on which is more important than the other, rather these are considered to be equally worthy as per WHO guidance. India is a large country with heterogeneous geographic, cultural and socio-economic features within and across regions. We therefore refrain from making an overarching comparison or ranking of the HAPs, in terms of the entirety of the data analysed. It is reiterated that this paper analyses the contents of the plans. If some component is in principle missing from the HAP, then that component is also unlikely to be implemented, and it seems reasonable to assume that what will be implemented is a subset of what is in the HAPs. From this perspective, as more city HAPs become available, similar analysis could be conducted on the components and sub-components of each plan, helping to identify and plug gaps for more effective implementation.

‘Agreement on a lead body,’ ‘health information plans,’ ‘accurate and timely alert systems’ and ‘reduction in indoor heat exposure’ find emphasis in the HAPs. Core elements that require long-term institutional and intersectoral planning and implementation such as ‘particular care for vulnerable groups,’ ‘preparedness of health/social care system,’ ‘long-term urban planning’ and ‘real-time surveillance’ need improvement. The findings indicate that praiseworthy efforts have been made in initiating HAPs for cities across India. The findings from these eight HAPs resonate with broader patterns identified in existing literature encompassing analysis of a wide array of HAPs²¹. This study complements these larger-scale assessments by offering more detailed, contextual and granular insights on heat action planning at the city level, highlighting thereby several opportunities to strengthen the plans.

The ontological and epistemological assumptions made regarding heat shape the policy and practice of heat management. Long-term adaptation and mitigation of extreme heat requires recognition of the specific characteristics of extreme heat, as distinct

from other extreme weather events²⁸. Two key findings emerge from the study. Firstly, the current approach of narrowly defining heat as a meteorological state, masks the recognition of subjective experiences of individuals and communities. Secondly, the continuing emphasis on extreme heat as a time-limited or acute disaster limits the scope of long-term policy planning which can reduce heat-health vulnerability. As a consequence, even when longer term measures are present, HAPs have a weaker effect on policy in comparison to emergency directives from higher levels of government due to weak institutionalisation and lack of dedicated resources²⁹.

While HAPs recognise the role of social vulnerability in adapting to heat, specific policy strategies to reduce vulnerability are required. Current solutions mostly emphasise individual action. Structural drivers of differential vulnerability need to be addressed²¹. The magnitude of heat is frequently characterised as the primary public health threat but explanations using solely objective heat measures, such as dew point temperature, relative humidity, or heat index, fail to explain and map differential impacts of heat³⁰. The health sector has a central role in setting guidelines, surveillance and evaluation for reducing heat-health risks^{31,32}.

Integrating the role of socio-economic status, age, caste, ethnicity, and gender in shaping heat-health relationships can lead to targeted strategies for impacted groups, improving their health to avoid both mortality from catastrophic events and long-term morbidities from sustained heat exposure. Operationalizing such an approach requires a community-centric focus with risk signatures, resource availability, and equitable access being mapped with community participation. Transitioning from a ‘community-placed’ to a ‘community-based’ approach, centered on community participation and knowledge³³ is possible. Community-based participatory research (CBPR) is a promising ‘bottom-up’ strategy where residents can map heat islands, identify vulnerable populations, and develop localised cooling solutions. Community-based campaigns using low-cost temperature sensors can further enhance urban heat island mapping and inform targeted interventions.

Community emergency response teams need training to assist specific vulnerable groups such as the elderly, disabled and children. The NDMA’s *Aapda Mitra* Scheme³⁴ which trains community volunteers for flood management, can potentially transform local

resilience during extreme heat events too. Climate resilience hubs, as locally managed neighbourhood centres that serve as safe havens during extreme heat events, can provide essential services like water, cool drinks, emergency supplies, and health information and shelters for the marginalised and homeless persons, for instance³⁵. By fostering community engagement and self-determination, these strategies can contribute to more effective and equitable heat management in urban areas.

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